

PROJECT EXPO

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Problem Statement

Road accidents are increasing rapidly worldwide, resulting in thousands of fatalities every year. One of the major reasons for these deaths is the delay in informing emergency medical services after an accident. In many cases, bystanders ignore the accident or fail to report it immediately. Existing accident detection systems mainly send an SMS after detecting an impact, but they often generate false alarms due to potholes, sudden braking, or minor collisions. Moreover, they do not verify the driver's condition before notifying emergency services. Therefore, there is a need for an intelligent accident detection system that can accurately identify major accidents, verify the driver's safety, and automatically notify family members and emergency responders with the live accident location.

Abstract

Road accidents are one of the leading causes of death worldwide, and delayed emergency response significantly increases the risk of fatalities. This project proposes an AI-Based Smart Vehicle Accident Detection and Emergency Rescue System using ESP32, MPU6050 accelerometer and gyroscope, vibration sensor, GPS, GSM/4G communication, and IoT technologies. The proposed system continuously monitors vehicle motion, acceleration, vibration, and tilt to accurately detect accidents while reducing false alarms caused by potholes or sudden braking.

When a severe accident is detected, the system activates a buzzer and automatically places a call to the driver. The driver is given sufficient time to respond by answering the call or pressing a cancel button. If the driver confirms safety, the emergency process is terminated. If there is no response within the specified time, the system automatically sends an SMS containing the live GPS location to registered family members and emergency contacts, followed by SOS emergency calls. The system also supports continuous live location updates until rescue personnel reach the accident site.

A backup battery ensures uninterrupted operation even if the vehicle's main power supply fails during the accident. An optional ESP32-CAM module can capture accident images for better emergency assessment. The proposed solution integrates IoT and AI techniques to improve accident detection accuracy and minimize false emergency alerts. This intelligent system aims to reduce emergency response time, increase the survival rate of accident victims, and provide a cost-effective and reliable safety solution for modern vehicles.

Objectives

- To detect vehicle accidents automatically using multiple sensors.
- To distinguish between minor and major accidents.
- To reduce false emergency alerts.
- To call the driver before contacting emergency services.
- To send live GPS location through SMS.
- To make SOS calls automatically when necessary.
- To provide continuous location updates.
- To improve emergency response time

Scope of the Project

The proposed system is designed for two-wheelers, cars, taxis, commercial vehicles, and public transport. It provides automatic accident detection, driver verification, GPS tracking, emergency notification, and IoT monitoring. The system can be expanded in the future using AI, cloud computing, and smart city infrastructure for enhanced road safety.

Expected Outcomes

Accurate accident detection.

Reduced false alarms.

Faster ambulance notification.

Live GPS tracking.

Automatic emergency calling.

Improved road safety.

Increased survival rate of accident victims.

LITERATURE SURVEY

Year	Author	<i>Research Work</i>
2024	Singh	IOT Vehicle Monitoring
2023	Patel	GSM and GPS-Based Accident Alert System
2024	Sharma	IoT-Based Vehicle Monitoring and Accident Detection

Existing Solutions Analysis

- Existing System
- Single impact sensor.
- Sends SMS immediately.
- No driver confirmation.
- No backup battery.
- No live GPS tracking.
- High false alarm rate.

Gap Analysis

Existing System	Proposed System
Single sensor	Multiple sensors
SMS only	SMS + SOS call
No driver verification	Driver confirmation
No backup battery	Backup battery included
No AI	AI-ready design
No live tracking	Continuous GPS tracking

Proposed Methodology

The system uses ESP32 as the controller. MPU6050, vibration sensor, and crash switch detect accident conditions. ESP32 analyzes sensor data and determines the severity of the impact. The driver is given time to respond. If no response is received, the system automatically sends GPS location through SMS, places SOS calls, and continues live GPS updates. A backup battery ensures operation during power failure.

References

- Espressif Systems, ESP32 Technical Reference Manual.
- InvenSense, MPU6050 Datasheet.
- SIM7600 Hardware Design Manual.
- H. Karl and A. Willig, Protocols and Architectures for Wireless Sensor Networks, Wiley.
- Simon Monk, Programming ESP32, McGraw-Hill.